
Math Training — M1 Cognitive Science

Session 4 — Analysis II

Linear systems, stability, oscillations, nonlinear dynamics

Advanced track

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Session 3 did functions with one input and one output. Almost nothing you will meet is like that.

- **Several inputs, one output:** $f(x, y)$ — a surface, its contour map, its *partial derivatives* and its *gradient*.
- **One input, several outputs:** $\mathbf{x}(t)$ — a trajectory, and its velocity.
- **Several of each:** the *Jacobian*.
- **Systems of differential equations:** $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, solved with the eigenvalues of Sessions 1–2.

■ If you remember only one thing

Everything today is the Session 3 derivative, applied **one variable at a time**. Nothing else is new. What *is* new is the bookkeeping: several derivatives must now be collected into a vector or a matrix, and that object is what carries the information.

So if a formula looks unfamiliar, look for the one-variable derivative inside it. It is always there.

■ Definition

A **function of several variables** takes a list of numbers and returns one number:

$$f : \mathbb{R}^n \longrightarrow \mathbb{R}, \quad (x_1, \dots, x_n) \longmapsto f(x_1, \dots, x_n).$$

For $n = 2$ its graph is a **surface**: the height $z = f(x, y)$ above each point of the plane.

Three examples, all of them things you already compute:

- accuracy as a function of task difficulty *and* amount of practice;
- a composite score $\mathbf{w} \cdot \mathbf{p}$ as a function of the weights;
- the total squared error of a fitted model, as a function of its parameters.

■ Definition

A **level set** (or contour line) of f is the set of points where f takes one fixed value: $\{(x, y) : f(x, y) = c\}$. Drawing several of them, for a ladder of values c , gives the **contour map**.

Read it exactly like a walking map. Contours close together mean a steep slope; contours far apart mean a flat region; a closed loop shrinking to a point is a summit or a basin.

■ Link to cognitive science

An *iso-performance* curve — all the combinations of difficulty and practice that give the same accuracy — is a level set. So is an indifference curve, and so is a confidence ellipse in a two-parameter fit.

■ Definition

The **partial derivative** $\frac{\partial f}{\partial x}$ measures how f changes when you nudge *only* x , holding every other variable fixed. So you differentiate as in Session 3, treating the other variables as constants.

Geometrically: slice the surface with a vertical plane in the x direction and take the slope of the curve you cut. One slope per direction.

■ Example

$$f(x, y) = x^2 + 3xy + y: \quad \frac{\partial f}{\partial x} = 2x + 3y, \quad \frac{\partial f}{\partial y} = 3x + 1.$$

Note that each partial derivative is still a *function of both variables* — the slope in the x direction generally depends on where you are in y too.

■ Definition

Collect the partial derivatives into a vector:

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right).$$

■ Key idea

∇f points in the direction of **steepest increase**, and its length is how steep that is. It is *perpendicular to the contour line* through the point — straight uphill is always across the contours, never along them.

The rate of change in an arbitrary unit direction $\mathbf{u}$ is the dot product $\nabla f \cdot \mathbf{u}$ — largest when $\mathbf{u}$ points along ∇f , and **zero** when $\mathbf{u}$ is along the contour. That is Session 1's projection picture, unchanged.

■ Link to cognitive science

Fit a straight line $\hat{y} = w_1x + w_0$ to n data points. The total squared error

$$J(w_0, w_1) = \sum_{i=1}^n (y_i - w_1x_i - w_0)^2$$

is a function of the two *parameters* — a bowl-shaped surface over the (w_0, w_1) plane. Fitting the model means finding the bottom of that bowl.

Two ways to get there, and both are on this slide:

- **Set the gradient to zero** and solve. For a linear model this is exact, and it is where the normal equations of least squares come from.
- **Walk downhill:** $\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla J(\mathbf{w})$, repeatedly. This is **gradient descent**, and it is how every model too complicated to solve exactly is fitted.

■ Definition

A **critical point** is a point where $\nabla f = \mathbf{0}$: every partial derivative vanishes, so the surface is flat there in all directions.

Three things it can be:

a **minimum**, a **maximum**, a **saddle**: uphill one way, downhill another.

The saddle is the genuinely new one — it has no analogue in one variable, and it is why “the gradient is zero” is not by itself good news when you are fitting a model.

For a bowl-shaped J like the least-squares error, there is exactly one critical point and it is the minimum. That is a real convenience, and it fails for most nonlinear models.

■ Practice — try it

1. For $f(x, y) = x^2 + 3xy + y$, compute both partial derivatives and $\nabla f(1, 2)$.
2. For $f(x, y) = x^2 + 4y^2$: sketch two contour lines, and give $\nabla f(1, 1)$. Check it is perpendicular to the contour.
3. For $f(x, y) = e^{-(x^2+y^2)}$, compute ∇f . Where is it zero, and is that a maximum or a minimum?
4. $J(w) = \sum_i (y_i - wx_i)^2$ with one parameter. Compute dJ/dw , set it to zero, and solve for w .

■ Definition

A **vector-valued function** of one variable returns a vector:

$$\mathbf{x} : \mathbb{R} \longrightarrow \mathbb{R}^n, \quad t \longmapsto \mathbf{x}(t) = (x_1(t), \dots, x_n(t)).$$

Its **derivative** is taken **componentwise**: $\dot{\mathbf{x}}(t) = (\dot{x}_1(t), \dots, \dot{x}_n(t))$, and it is again a vector.

That is the whole definition. There is no new rule — n ordinary derivatives, stacked.

Geometrically, $\mathbf{x}(t)$ traces a **curve** as t runs, and $\dot{\mathbf{x}}(t)$ is the **velocity**: it points along the curve, in the direction of travel, and its length is the speed.

■ Link to cognitive science

In a mouse-tracking experiment the participant moves the cursor from a start button to one of two response buttons, and the position is recorded every few milliseconds:

$$\mathbf{x}(t) = (x(t), y(t)).$$

A trajectory that curves toward the unchosen option before committing is the standard evidence that the two responses were competing — and “curves toward” is a statement about the direction of $\dot{\mathbf{x}}$.

Eye position over time, a two-joint reaching movement, and the pair (accuracy, speed) of a participant across a session are all the same kind of object.

Note the contrast with Session 1: there, a participant was *one* point. Here a participant is a whole *curve*.

■ Example

$$\mathbf{x}(t) = (t, t^2) \quad \text{and} \quad \mathbf{y}(t) = (\cos t, \sin t).$$

First: $\dot{\mathbf{x}}(t) = (1, 2t)$. The path is the parabola $y = x^2$; at $t = 0$ the velocity is $(1, 0)$ — moving horizontally — and at $t = 1$ it is $(1, 2)$, climbing steeply. Speed $\|\dot{\mathbf{x}}\| = \sqrt{1 + 4t^2}$ increases with t .

Second: $\dot{\mathbf{y}}(t) = (-\sin t, \cos t)$. The path is the unit circle. Note that

$$\mathbf{y} \cdot \dot{\mathbf{y}} = -\cos t \sin t + \sin t \cos t = 0 :$$

the velocity is always perpendicular to the position. Constant speed $\|\dot{\mathbf{y}}\| = 1$, constant distance from the origin — which is what going round in a circle means.

■ Definition

For $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with components $F_1, \dots, F_m$, the **Jacobian** is the $m \times n$ matrix of all the partial derivatives:

$$\mathbf{J} = \begin{pmatrix} \partial F_1 / \partial x_1 & \cdots & \partial F_1 / \partial x_n \\ \vdots & & \vdots \\ \partial F_m / \partial x_1 & \cdots & \partial F_m / \partial x_n \end{pmatrix}.$$

Row i is the gradient of F_i .

■ Key idea

$\mathbf{J}$ is the **best linear approximation** of $\mathbf{F}$ near a point: $\mathbf{F}(\mathbf{x}_0 + \delta) \approx \mathbf{F}(\mathbf{x}_0) + \mathbf{J} \delta$.
That is the one-variable tangent line, with a matrix in place of the slope.

■ Example

$\mathbf{F}(x, y) = (x - x^3 - y, x - y)$. Compute $\mathbf{J}$, then evaluate at $(0, 0)$ and at $(1, 1)$.

Row by row:

$$\mathbf{J}(x, y) = \begin{pmatrix} 1 - 3x^2 & -1 \\ 1 & -1 \end{pmatrix}.$$

At the origin, $\mathbf{J}(0, 0) = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$; at $(1, 1)$, $\mathbf{J}(1, 1) = \begin{pmatrix} -2 & -1 \\ 1 & -1 \end{pmatrix}$.

Two things to notice. The Jacobian is a *function of position* — it is only constant when $\mathbf{F}$ is linear. And the entry $J_{12} = -1$ answers a specific question: if I nudge y upward, F_1 goes down at unit rate.

■ Practice — try it

1. $\mathbf{x}(t) = (2t, 3 - t)$: compute $\dot{\mathbf{x}}$, the speed, and describe the path.
2. $\mathbf{x}(t) = (e^{-t} \cos t, e^{-t} \sin t)$: compute $\|\mathbf{x}(t)\|$ and say in words what the path does.
3. Write the Jacobian of $\mathbf{F}(x, y) = (-x + y, 2x - 3y)$. Why is it constant?
4. Write the Jacobian of $\mathbf{F}(x, y) = (x^2y, x + \sin y)$ and evaluate it at $(1, 0)$.

Last session: one quantity, $\dot{x} = ax + b$. But quantities usually influence each other, and then you get one equation per quantity:

$$\dot{x}_1 = a_{11}x_1 + a_{12}x_2, \quad \dot{x}_2 = a_{21}x_1 + a_{22}x_2,$$

which is exactly

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}.$$

■ Key idea

The unknown is now a **vector-valued function** $\mathbf{x}(t)$ — part B — and the coefficients are a **matrix** — Sessions 1–2. The whole of the rest of the session is those two things meeting.

■ Definition

Change coordinates to the eigenvectors of $\mathbf{A}$. Along an eigendirection the system is just $\dot{c} = \lambda c$ — a Session 3 equation — so

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2,$$

with c_1, c_2 fixed by the initial condition $\mathbf{x}(0)$.

Read it as a recipe:

1. find the eigenvalues and eigenvectors of $\mathbf{A}$ (Session 2, one quadratic);
2. write down one exponential per eigenvalue;
3. fix the two constants from $\mathbf{x}(0)$.

A two-dimensional problem has become two independent one-dimensional ones.

■ Key idea

$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ decays to zero **if and only if** $\operatorname{Re} \lambda < 0$ for *every* eigenvalue of $\mathbf{A}$.

One eigenvalue with positive real part destroys stability, however negative the others are — there is no averaging. And the inequality is strict: an eigenvalue sitting exactly on the imaginary axis settles nothing.

This is the one-variable rule of Session 3 ($a < 0$) with two changes: there is now one condition per eigenvalue, and “negative” has become “negative *real part*”, because eigenvalues may be complex.

■ Useful fact

- real, same sign $\rightarrow$ **node** (stable if both < 0 , unstable if both > 0);
- real, opposite signs $\rightarrow$ **saddle** — always unstable, but stable along one direction;
- complex $\mu \pm i\omega$ with $\mu \neq 0 \rightarrow$ **spiral**, inward if $\mu < 0$;
- purely imaginary $\rightarrow$ **centre**: closed orbits, neither growing nor decaying.

And you can read all of this off $T = \text{tr } \mathbf{A}$ and $D = \det \mathbf{A}$ without ever computing an eigenvalue:

$$D < 0 \Rightarrow \text{saddle}, \quad D > 0, T < 0 \Rightarrow \text{stable}, \quad D > 0, T > 0 \Rightarrow \text{unstable},$$

$$T^2 < 4D \Rightarrow \text{spiral}, \quad T^2 > 4D \Rightarrow \text{node}.$$

■ Example

Classify and solve $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ for $\mathbf{A} = \begin{pmatrix} -1 & -2 \\ 2 & -1 \end{pmatrix}$.

$T = -2$ and $D = 1 + 4 = 5$, so $\lambda^2 + 2\lambda + 5 = 0$ with discriminant $4 - 20 = -16 < 0$.
Hence

$$\lambda = -1 \pm 2i.$$

The real part -1 is negative, so trajectories decay; the imaginary part 2 is nonzero, so they turn. This is a **stable spiral**: the state circles at angular speed 2 while its distance from the origin shrinks like e^{-t} .

Check against T and D : $D = 5 > 0$ and $T = -2 < 0$, so stable; and $T^2 = 4 < 4D = 20$, so a spiral. Same answer, no complex arithmetic.

Compare part B, exercise 2: $(e^{-t} \cos t, e^{-t} \sin t)$ was this picture, written down directly.

■ Link to cognitive science

A standard model of choice: two evidence accumulators, x for option A and y for option B, each decaying and each *inhibiting* the other with strength $c > 0$:

$$\dot{x} = -x - cy, \quad \dot{y} = -cx - y, \quad \mathbf{A} = \begin{pmatrix} -1 & -c \\ -c & -1 \end{pmatrix}.$$

This matrix is symmetric, so Session 2 hands us its eigenvectors immediately:

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1) \text{ with } \lambda_1 = -1 - c, \quad \mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1) \text{ with } \lambda_2 = -1 + c.$$

The *total* activity $x + y$ always decays. The *difference* $x - y$ decays if $c < 1$ — the two options stay balanced — and **grows** if $c > 1$: a small initial advantage is amplified until one option wins. Strong mutual inhibition turns a balanced state into a saddle, and that is a decision.

■ Definition

For $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$, a **fixed point** $\mathbf{x}^*$ is where $\mathbf{F}(\mathbf{x}^*) = \mathbf{0}$: nothing changes there. Near it, write $\delta = \mathbf{x} - \mathbf{x}^*$; then

$$\dot{\delta} \approx \mathbf{J}(\mathbf{x}^*) \delta,$$

so the local behaviour is decided by the eigenvalues of the **Jacobian evaluated at that point**.

That is the entire method, and every ingredient is already on the previous slides: part A gave the partial derivatives, part B assembled them into $\mathbf{J}$, and part C classifies the resulting matrix.

The approximation is trustworthy as long as no eigenvalue has zero real part.

■ Definition

For $\dot{x} = f(x, y)$ and $\dot{y} = g(x, y)$, the **nullclines** are the curves $f = 0$ (where x momentarily stops changing) and $g = 0$ (where y does). **Fixed points sit where the two nullclines cross.**

Draw them first, always. They cost nothing, they locate every fixed point at once, and they already tell you the direction of flow in each region between them — before a single derivative has been computed.

Then, at each crossing: compute $\mathbf{J}$, get T and D , and read off the type.

■ Example

Analyse $\dot{x} = x - x^3 - y$, $\dot{y} = x - y$.

Nullclines. $\dot{y} = 0$ gives $y = x$. Substituting into $\dot{x} = 0$: $x - x^3 - x = 0$, i.e. $x^3 = 0$, so the only fixed point is the origin.

Jacobian. From the worked example in part B, $\mathbf{J}(0,0) = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$.

Classify. $T = 1 - 1 = 0$ and $D = (1)(-1) - (-1)(1) = 0$. Both are zero, so *both* eigenvalues are zero and the linearisation decides nothing at all.

That is an honest outcome, not a failure — and it is exactly the case the grey caveat warned about. When it happens, the nullcline picture and the direction of flow are what you fall back on.

■ Practice — try it

For each matrix give T , D , the eigenvalues, and the type of the flow.

1. $\begin{pmatrix} -2 & 0 \\ 0 & -5 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 0 & -3 \end{pmatrix}$.
2. $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.
3. A matrix with eigenvalues $-0.1 \pm 3i$: stable? oscillating? at roughly what angular speed?
4. For $\dot{x} = y$, $\dot{y} = -x - y$: find the fixed point, write $\mathbf{J}$, and classify it.
5. In the two-accumulator model, take $c = 2$. Give both eigenvalues and say which direction grows.

- **Several inputs, one output:** contour maps, *partial derivatives*, and the *gradient* — which points straight uphill, is perpendicular to the contours, and is how every model gets fitted.
- **One input, several outputs:** a *trajectory*, differentiated componentwise to give the *velocity*.
- **Several of each:** the *Jacobian*, the best linear approximation — the tangent line with a matrix in place of the slope.
- **Systems:** $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ solves as one exponential per eigenvector; stable exactly when every $\text{Re } \lambda < 0$; node, saddle, spiral or centre, read off T and D . Nonlinear systems are handled by linearising at each fixed point.

You can now read the phase portraits and the fitting procedures that appear throughout the M2 as routine calculations.